

Internet of Things

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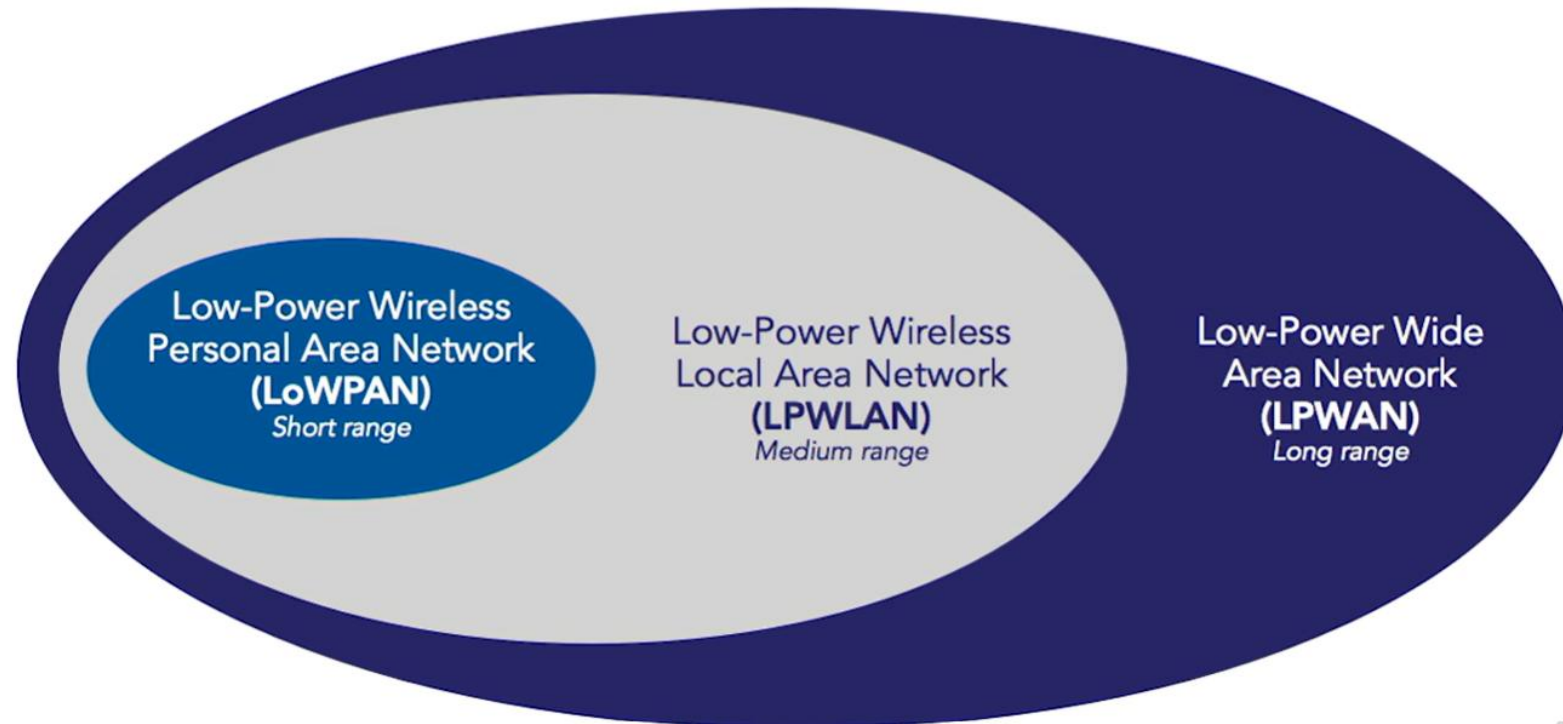
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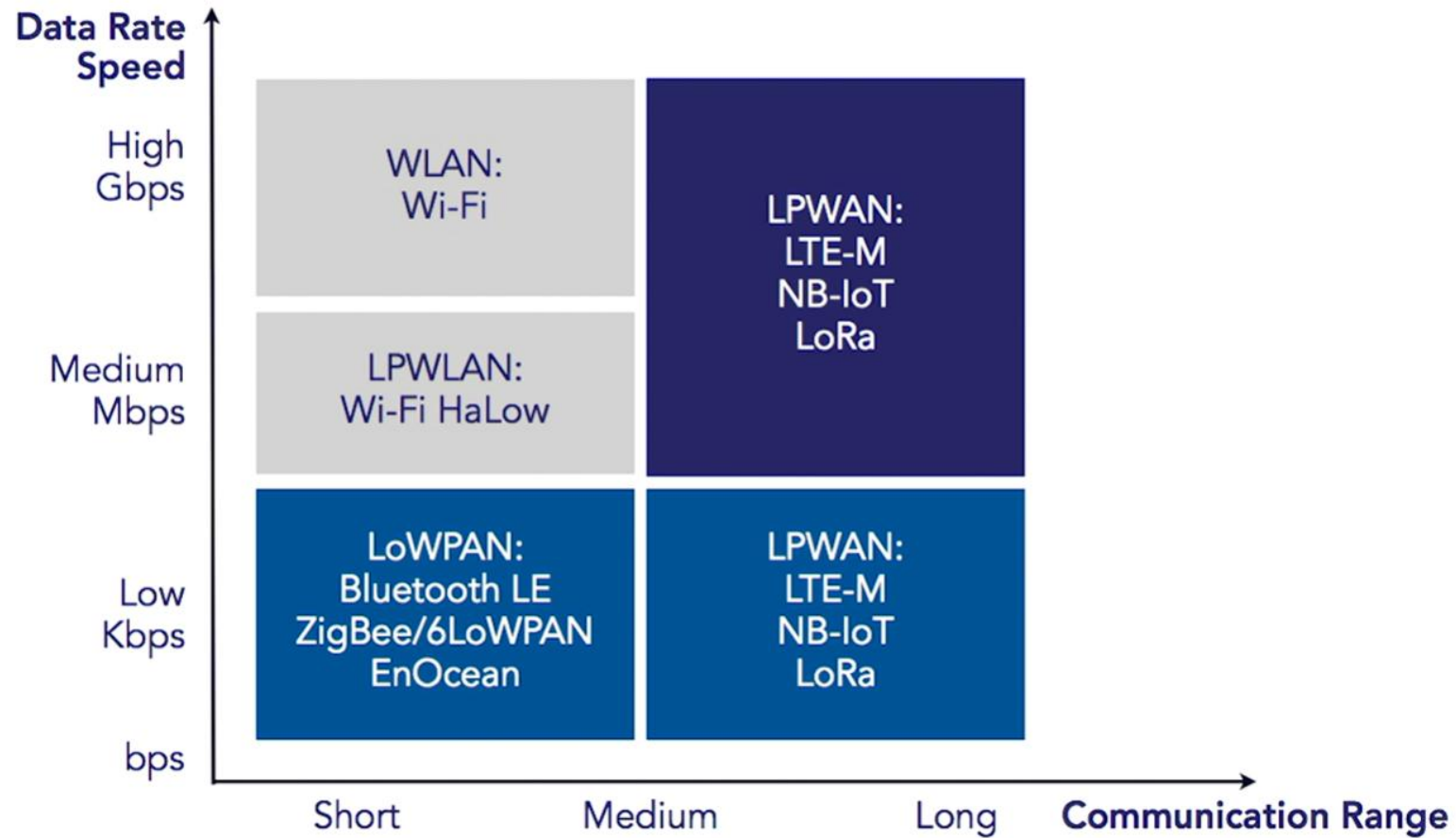
Outline

- ▶ IoT Foundations: Low-Power Wireless Networking
- ▶ IoT network topologies and Architectures
- ▶ IoT system Architecture and Elements
- ▶ Data Communication Paradigms for IoT
- ▶ Data communication in IoT elements
- ▶ Performance metrics
- ▶ Performance measurement
- ▶ Energy harvesting based IoT devices

Low-Power Wireless Network Types



Low-Power Wireless Networking



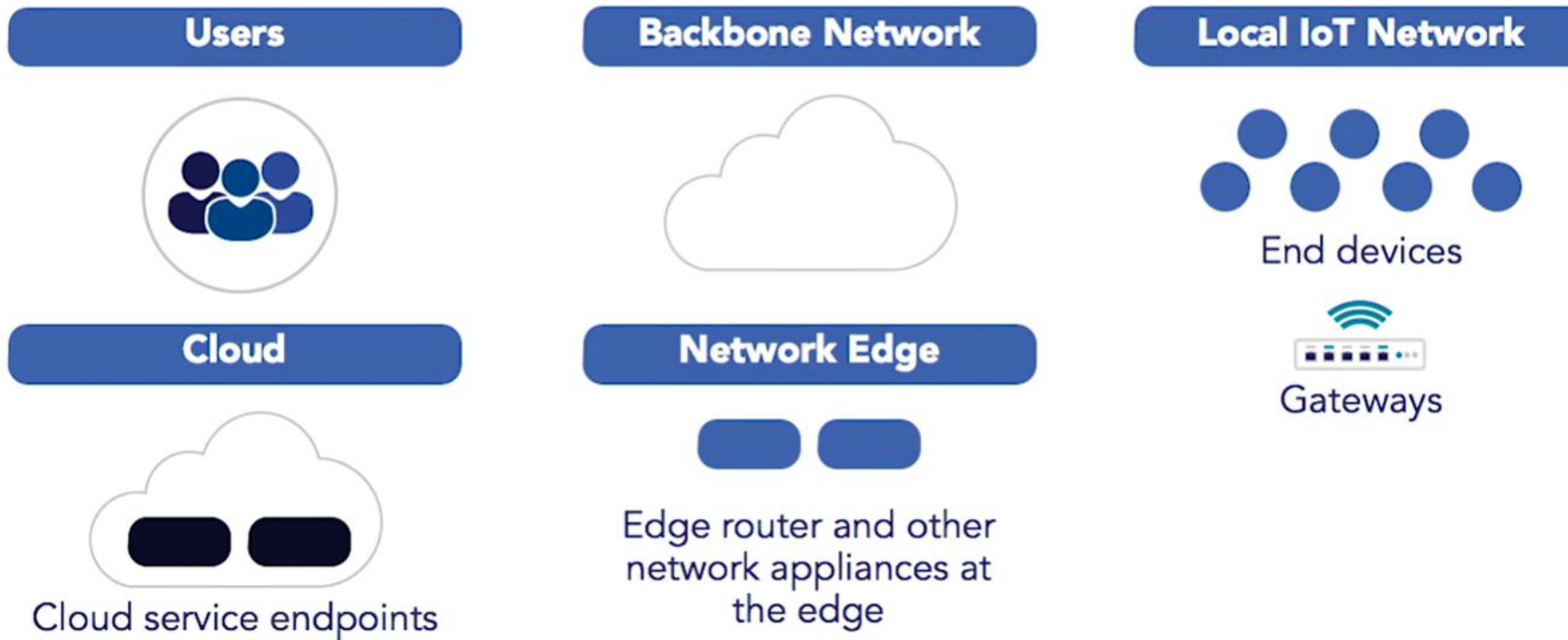
Examples of LoWPAN Technologies

- Bluetooth Low Energy
- ZigBee
- 6LoWPAN
- WirelessHART

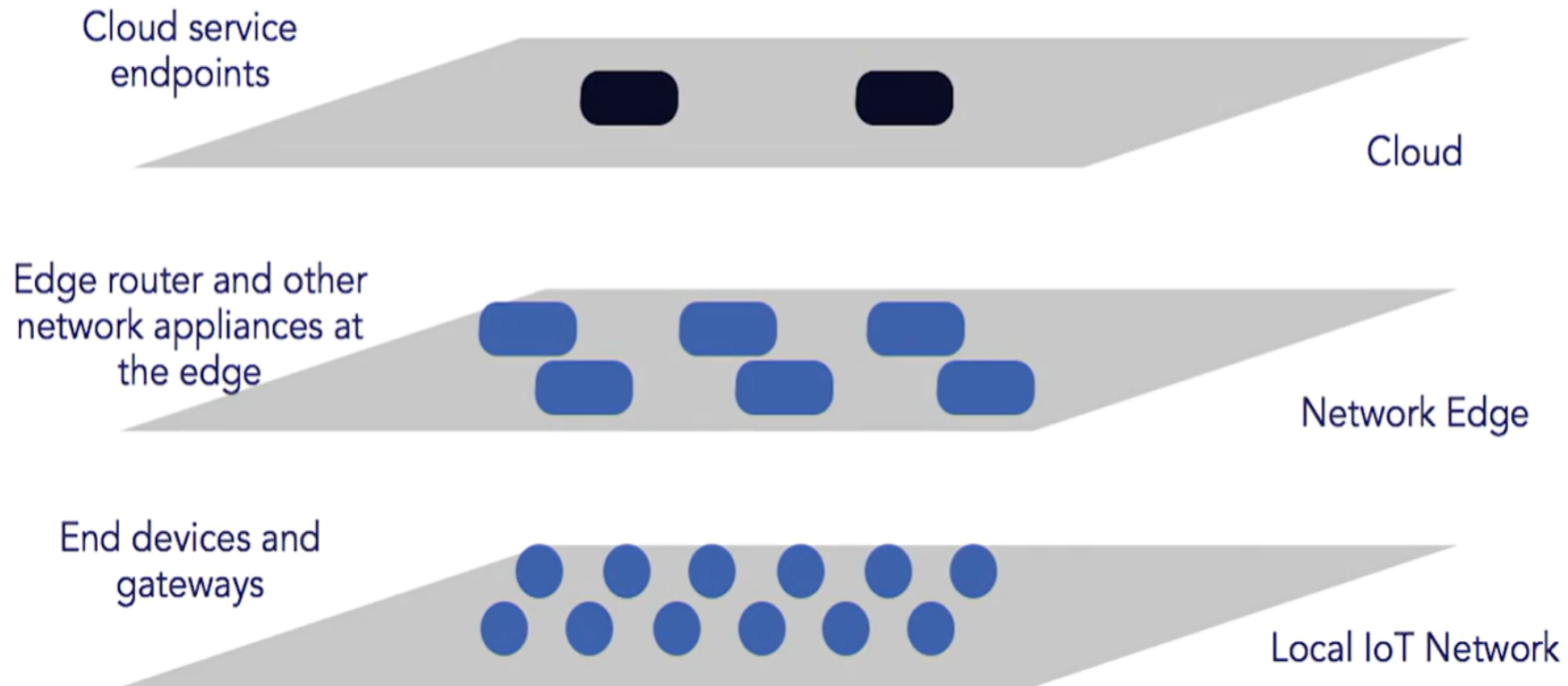
Examples of LPWAN Technologies

- LTE-M (Long term evolution for machine type communication)
- NB-IoT (Narrowband IoT)
- EC-GSM-IoT (Extended coverage GSM IoT)
- LoRa
- SIGFOX

Elements in IoT system

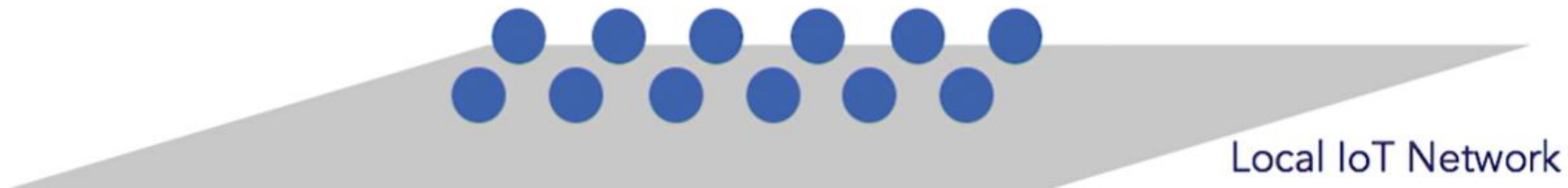


Elements in IoT system



Local IoT network

- End devices are the sources of machine data.
- End devices can generate big sensing data over time.
- End devices may need to handle control data.



Computing on the Local IoT network

Local data processing

End devices make local processing.

Collaborative data processing

End devices exchange the data and process them collaboratively.

Central data processing

End devices can transmit the data directly to the remote application service.

Computing at the edge

- Handle the data from the local IoT network
- Transfer data to the backbone network
- Data processing at the edge can reduce data traffic on the backbone network and at the cloud.

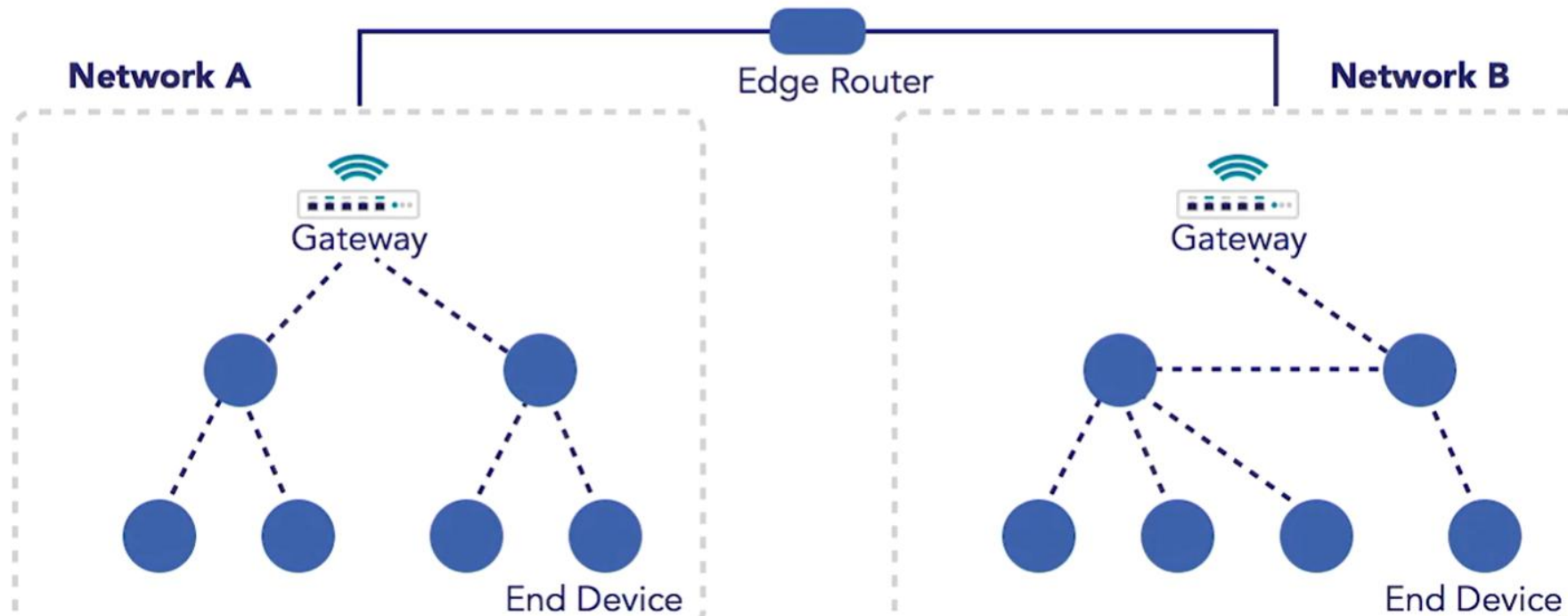
Computing at the cloud

- A cloud service can receive from a local IoT network and send data to it.

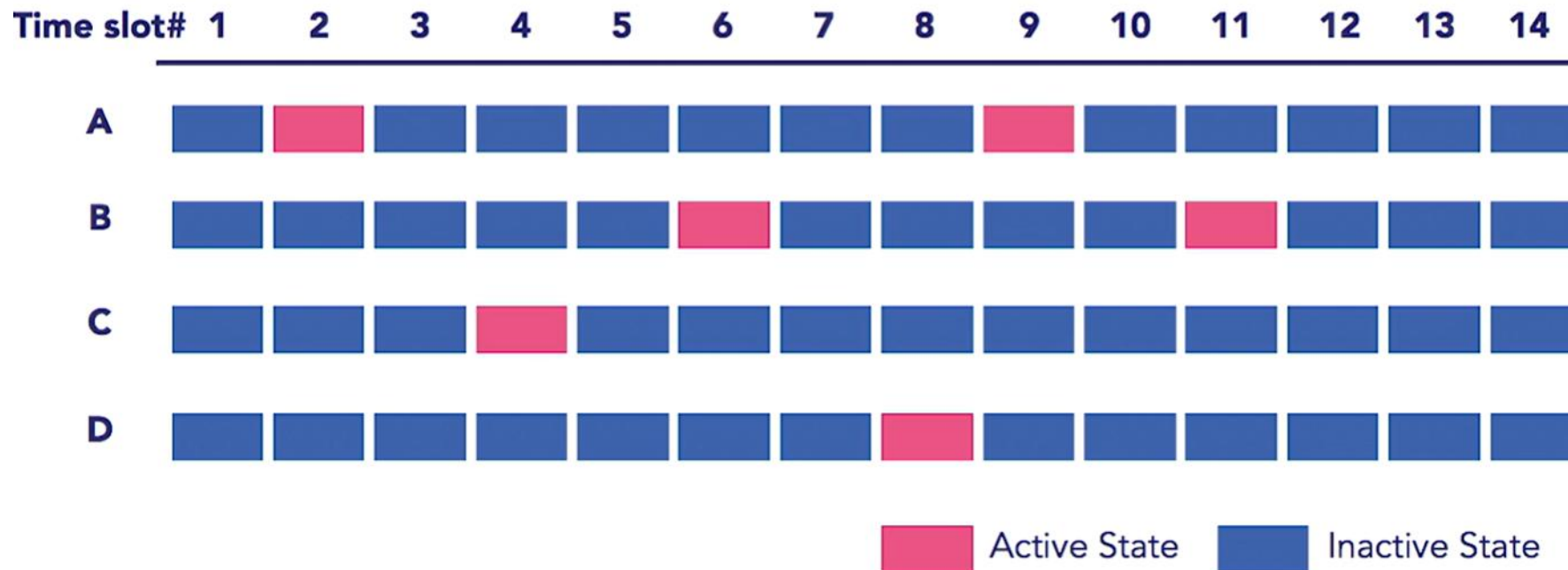
Data communication in a Local Network

- End devices can communicate with each other.
- End devices only communicate with a gateway/edge router.
- End devices communicate with each other and with a gateway/edge router.

Data communication in a Local Network



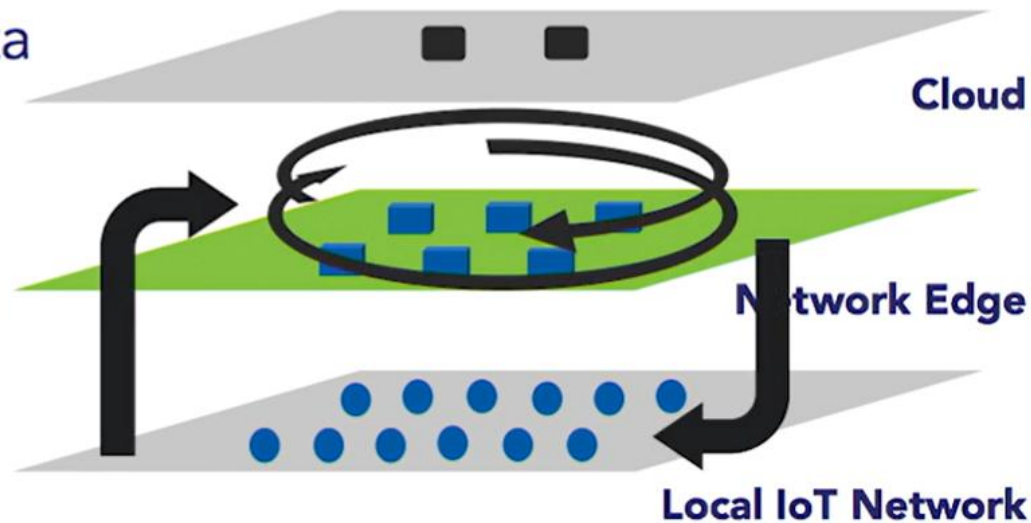
Duty Cycle based communication



Only transmit data in active state

Data communication at the Edge

- Make an intelligent decision for the local network
- Process the machine data and send to a cloud
- Directly transmit the machine data to a cloud



Communication between cloud and local network

Process and manage machine data

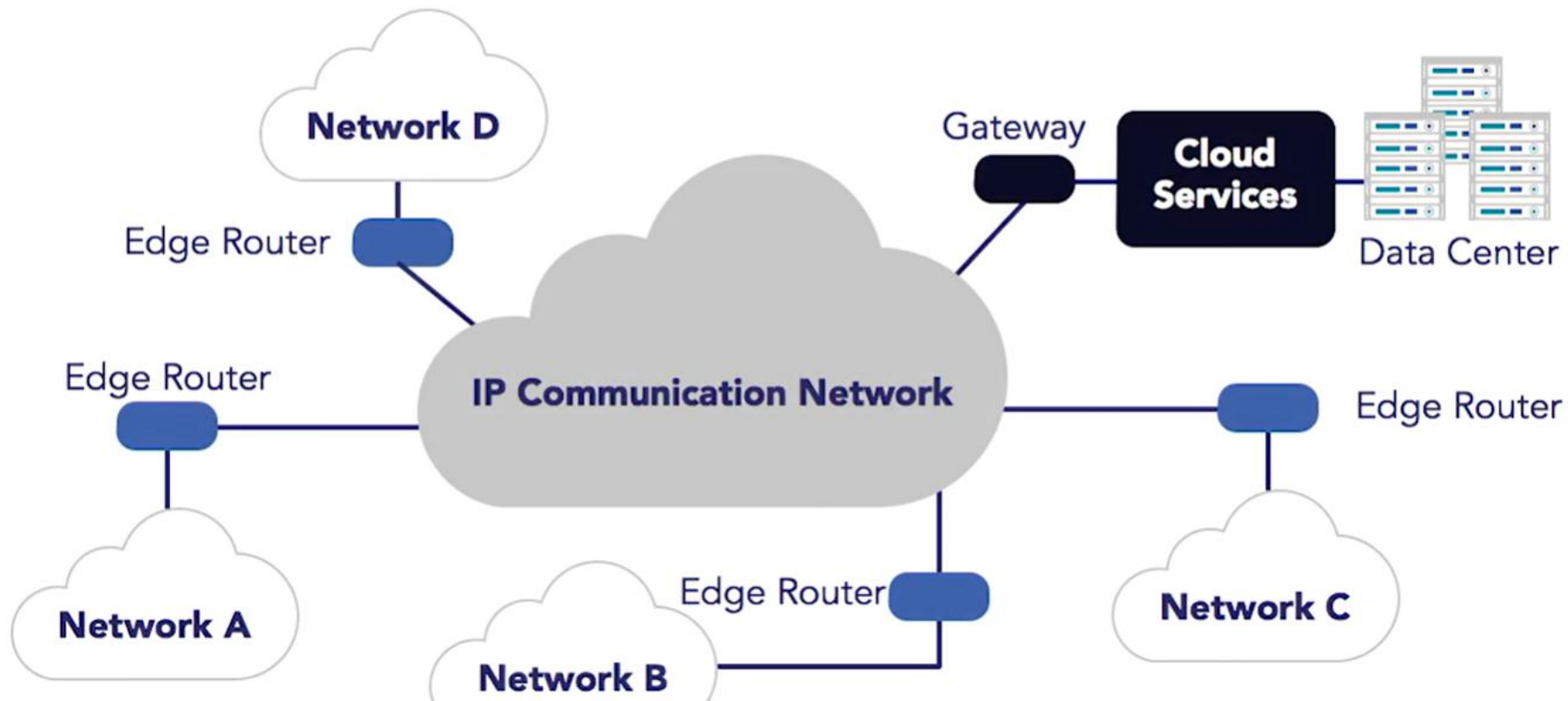
Security audit, data storage, visualization, analysis, etc.

Interact with users

Data and/or device management, system configurations, etc.

Communicate with distributed local networks

Communication between cloud and local network



Performance metrics

- Bit rate
- Throughput
- Latency
- Packet delay variation
- Packet loss

Bit rate

It is defined as the number of bits transmitted per second.
It is often expressed as bps, or bit/s.

kbps (or kbit/s): 1,000 bits per second

Mbps (or Mbit/s): 1,000,000 bits per second

Gbps (or Gbit/s): 1,000,000,000 per second

Throughput

Network throughput

The amount of data transmitted per unit of time over a communication path

Application-level throughput (or goodput)

The amount of application data transmitted per unit of time over a communication path

Latency

- It is measured as the amount of time taken to transmit a packet between a sender and a receiver.
- It involves the delay caused by several factors, such as packet processing, transmissions, signal propagation, intermediate network devices, etc.

One Way delay

Data Packet

0110101011101010



Two Way delay

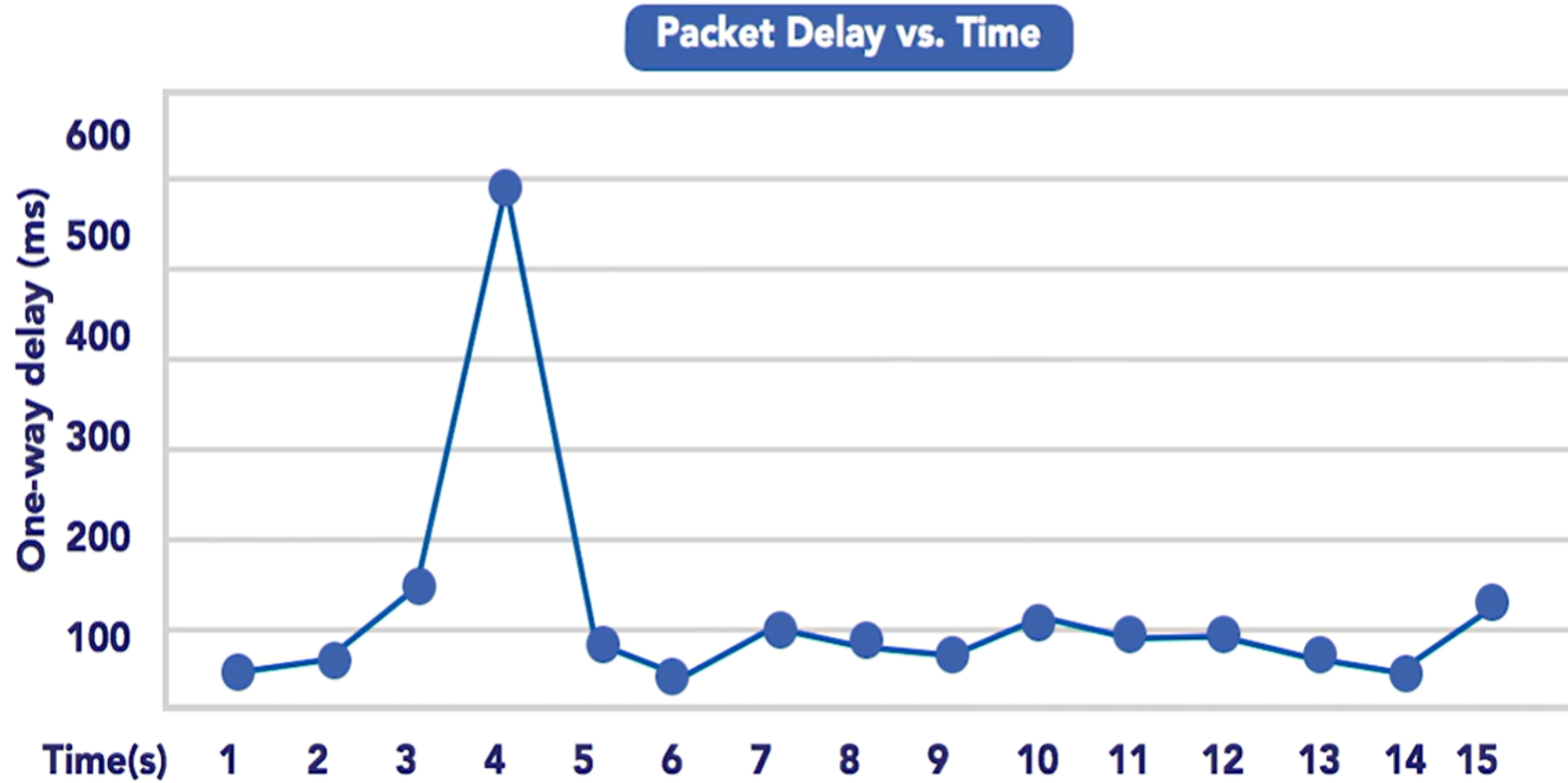
Data Packet

0110101011101010



0110101011101010

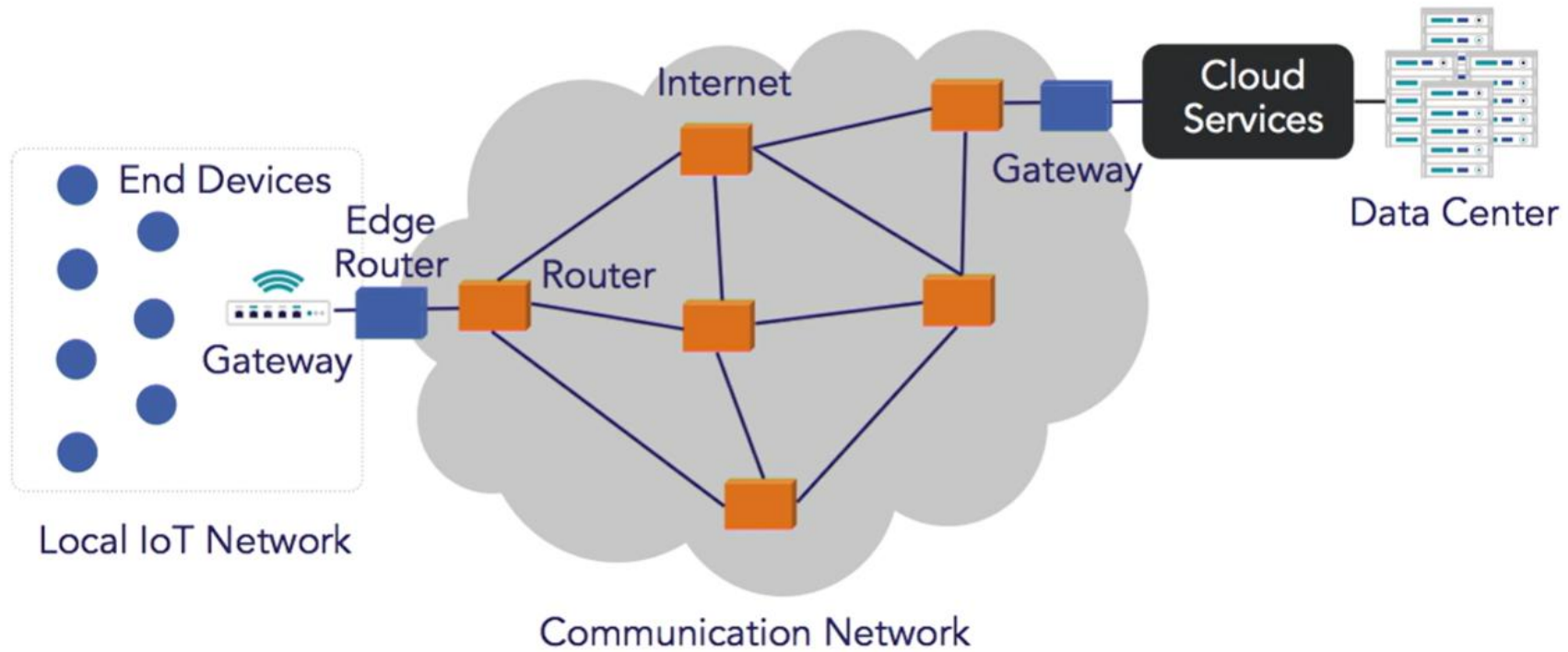
Packet delay variation



Packet loss

Packet loss occurs when a packet travels through a network. It is measured as a percentage of packets lost with regard to packets sent.

Performance measurement



Spectrum Analyzer



- Display the RF signal's amplitude as a function of frequency and time.
- Measure frequency , power , modulation, noise etc.

Packet Analyzer/ Sniffer

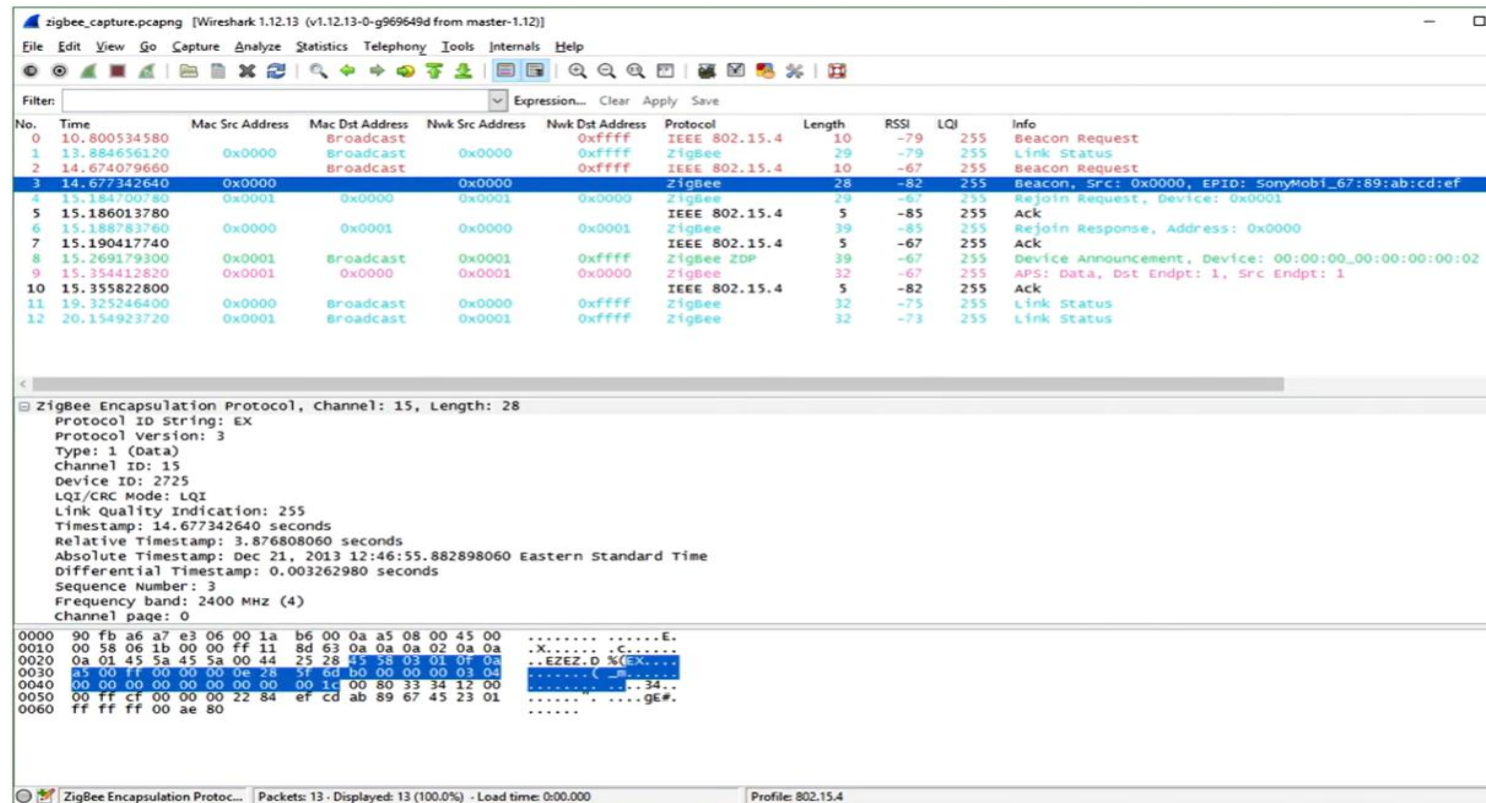


CC2540EMK-USB Dongle

Radio front end (e.g. Bluetooth LE/IEEE 802.15.4 USB dongle)

Packet analysis software (e.g. Wireshark)

Zigbee packet capture with Wireshark



The image shows a Wireshark packet capture of Zigbee traffic. The main packet list table displays 13 captured packets. Packet 3 is selected, showing its details in the packet details pane and its raw bytes in the packet bytes pane.

No.	Time	Mac Src Address	Mac Dst Address	Nwk Src Address	Nwk Dst Address	Protocol	Length	RSSI	LQI	Info
0	10.800534580		Broadcast		0xffff	IEEE 802.15.4	10	-79	255	Beacon Request
1	13.884656120	0x0000	Broadcast	0x0000	0xffff	ZigBee	29	-79	255	Link Status
2	14.674079660		Broadcast		0xffff	IEEE 802.15.4	10	-67	255	Beacon Request
3	14.677342640	0x0000		0x0000		ZigBee	28	-82	255	Beacon, Src: 0x0000, EPID: SonyMobi_67:89:ab:cd:ef
4	15.186700780	0x0001	0x0000	0x0001	0x0000	ZigBee	29	-85	255	Rejoin Request, Device: 0x0001
5	15.186013780					IEEE 802.15.4	5	-85	255	Ack
6	15.188783760	0x0000	0x0001	0x0000	0x0001	ZigBee	39	-85	255	Rejoin Response, Address: 0x0000
7	15.190417740					IEEE 802.15.4	5	-67	255	Ack
8	15.269179300	0x0001	Broadcast	0x0001	0xffff	ZigBee ZDP	39	-67	255	Device Announcement, Device: 00:00:00_00:00:00:00:02
9	15.354412820	0x0001	0x0000	0x0001	0x0000	ZigBee	32	-67	255	APS: Data, Dst Endpt: 1, Src Endpt: 1
10	15.355822800					IEEE 802.15.4	5	-82	255	Ack
11	19.325246400	0x0000	Broadcast	0x0000	0xffff	ZigBee	32	-75	255	Link Status
12	20.154923720	0x0001	Broadcast	0x0001	0xffff	ZigBee	32	-73	255	Link Status

Packet 3 Details:

- ZigBee Encapsulation Protocol, Channel: 15, Length: 28
- Protocol ID string: EX
- Protocol version: 3
- Type: 1 (Data)
- Channel ID: 15
- Device ID: 2725
- LQI/CRC Mode: LQI
- Link Quality Indication: 255
- Timestamp: 14.677342640 seconds
- Relative Timestamp: 3.876808060 seconds
- Absolute Timestamp: Dec 21, 2013 12:46:55.882898060 Eastern Standard Time
- Differential Timestamp: 0.003262980 seconds
- Sequence Number: 3
- Frequency band: 2400 MHz (4)
- Channel page: 0

Packet 3 Raw Bytes:

```
0000  90 fb a6 a7 e3 06 00 1a b6 00 0a a5 08 00 45 00 .....E.
0010  00 58 06 1b 00 00 ff 11 8d 63 0a 0a 0a 02 0a 0a ..X.....C.
0020  0a 01 45 5a 45 5a 00 44 25 28 15 15 03 01 01 02 ..EZZ.D %()
0030  a5 00 ff 00 00 00 0e 28 5f 8d b0 00 00 00 03 04 .....(..#...
0040  00 00 00 00 00 00 00 00 00 1c 00 80 33 34 12 00 .....34..
0050  00 ff cf 00 00 00 22 84 ef cd ab 89 67 45 23 01 .....GE#.
0060  ff ff ff 00 ae 80 .....
```

Zigbee packet capture with Wireshark

The image displays the Wireshark network protocol analyzer interface, specifically showing a capture of Zigbee packets. The interface is divided into three main panes: the Packet List Pane at the top, the Packet Detail Pane in the middle, and the Packet Bytes Pane at the bottom.

Packet List Pane: This pane shows a list of captured packets. The columns include No., Time, Mac Src Address, Mac Dst Address, Nwk Src Address, Nwk Dst Address, Protocol, Length, RSSI, LQI, and Info. The packets are filtered by 'Expression... Clear Apply Save'. The list shows 13 packets, with the 3rd packet (No. 3) selected, which is a ZigBee Beacon Request.

Packet Detail Pane: This pane provides detailed information about the selected packet (No. 3). It shows the ZigBee Encapsulation Protocol details, including Protocol ID String: EX, Protocol Version: 3, Type: 1 (Data), Channel ID: 15, Device ID: 2725, LQI/CRC Mode: LQI, Link Quality Indication: 255, Timestamp: 14.677342640 seconds, Relative Timestamp: 3.876808060 seconds, Absolute Timestamp: Dec 21, 2013 12:46:55.882898060 Eastern Standard Time, Differential Timestamp: 0.003262980 seconds, Sequence Number: 3, Frequency band: 2400 MHz (4), and channel page: 0.

Packet Bytes Pane: This pane shows the raw bytes of the selected packet. It displays the packet structure in hexadecimal and ASCII. The packet is a ZigBee Beacon Request, and the bytes are shown in a hex dump format.

Texas Instruments SmartRF Packet Sniffer Bluetooth Low Energy

File Settings Help

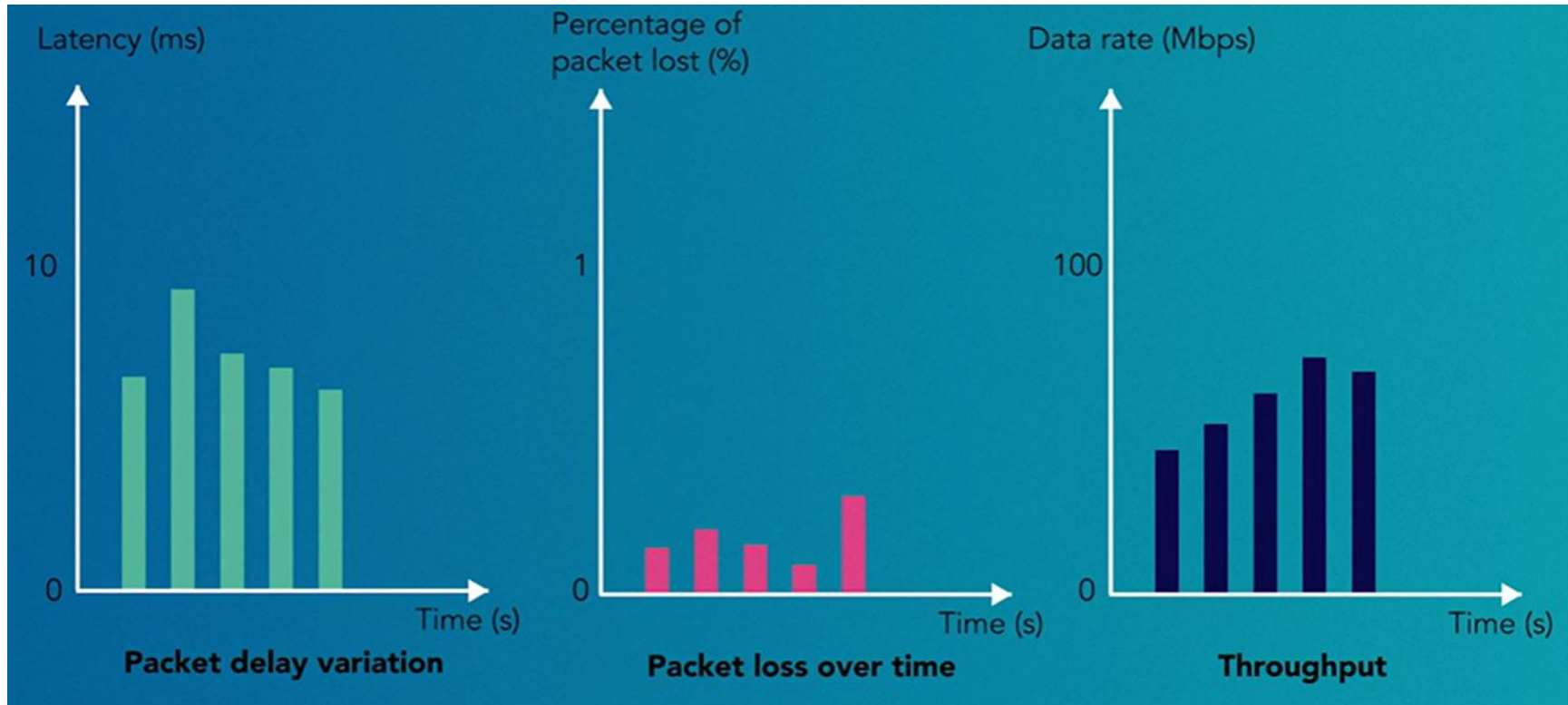
Capturing device | Radio Configuration | Select fields | Packet details | Address book | Display filter | Time line

Packet count: 853	Error count: 34	Filter off	RF device: CC2540	Channel: 37 [0x25]	Packet broadcast OFF
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Performance metric

- Determine the performance metric to measure the variance of the data rate transferred over time

Performance metric



Energy Harvesting based Wireless Network technologies

- ▶ The process in which energy is captured from the ambient environment and converted into usable electric power.

Energy Harvesting based Wireless Network technologies

- ▶ Application
 - ▶ Home automation /smart home
 - ▶ Building automation
 - ▶ General M2M application

Network supporting EH devices

- Proprietary EH-based networking technologies
- EnOcean
- ZigBee PRO with Green Power
- Bluetooth LE

Smart home application



Lighting System



Motion-Powered Switch



HVAC System

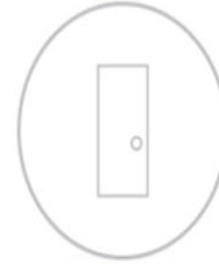


Solar-Powered Occupancy Sensor

Solar-Powered Temperature
and Humidity Sensor

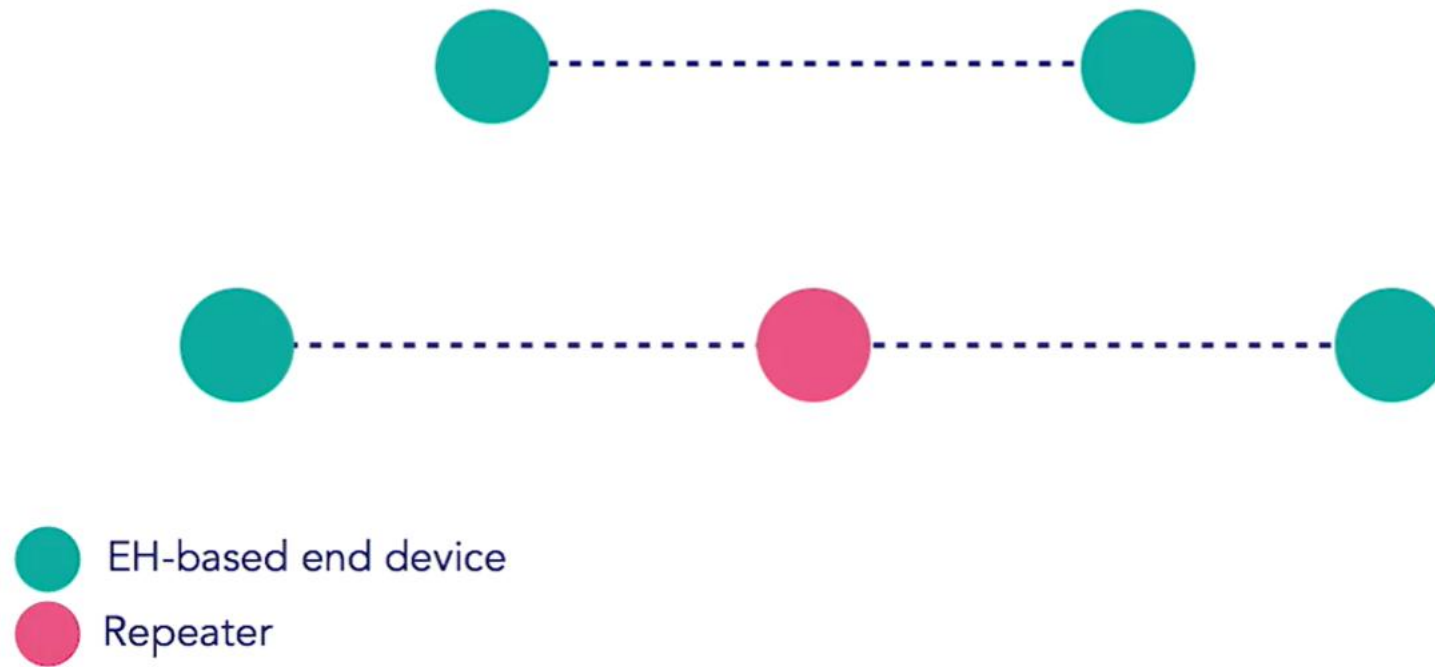


Window/Door Monitoring

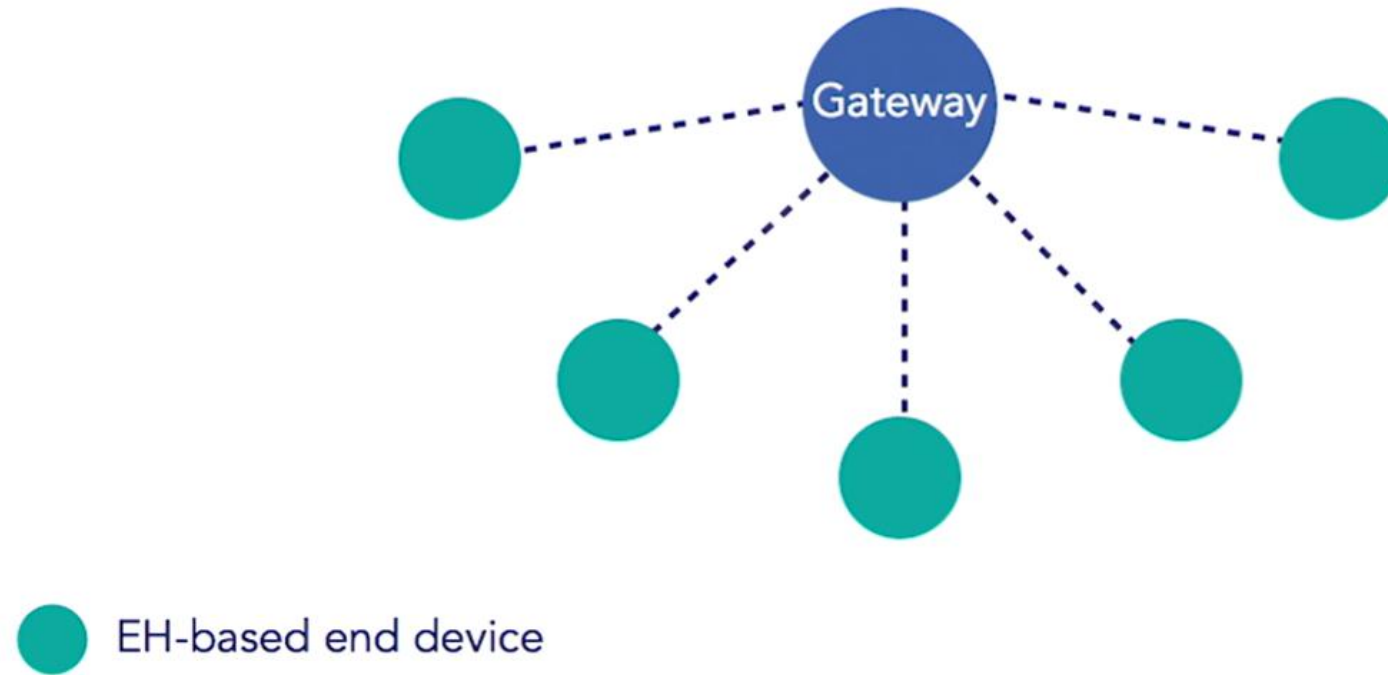


Motion-Powered Window/Door Handle

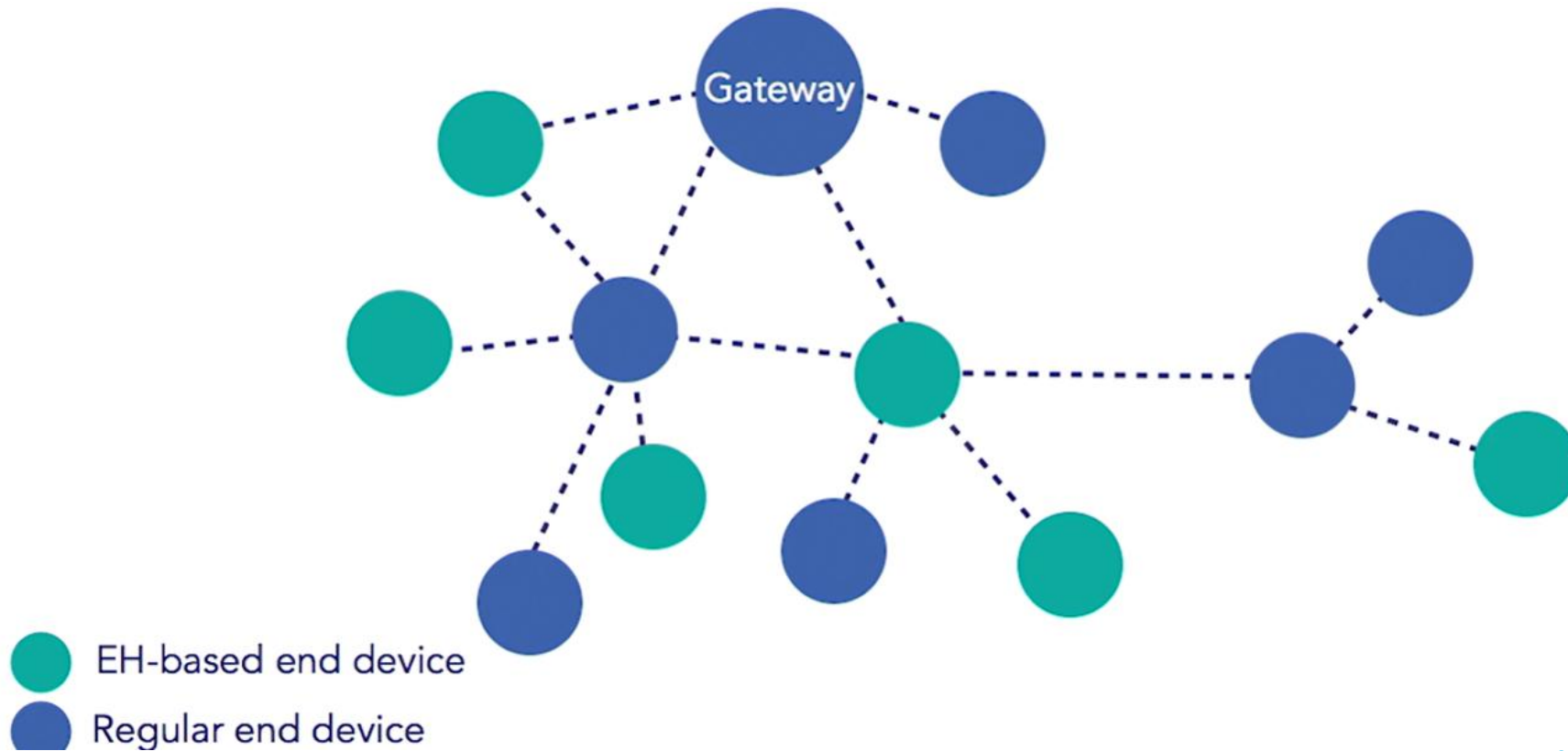
Point to point Topology



Star Topology based on EH End devices



Mesh Topology Based on EH End devices



Energy Harvesting Module



Energy harvesting (EH, also known as power harvesting) is the process by which energy is derived from external sources (e.g., solar power, thermal energy, wind energy, salinity gradients, and kinetic energy, also known as ambient energy), and then stored for use by small, wireless autonomous devices, like those used in wearable electronics, condition monitoring, and wireless sensor networks.

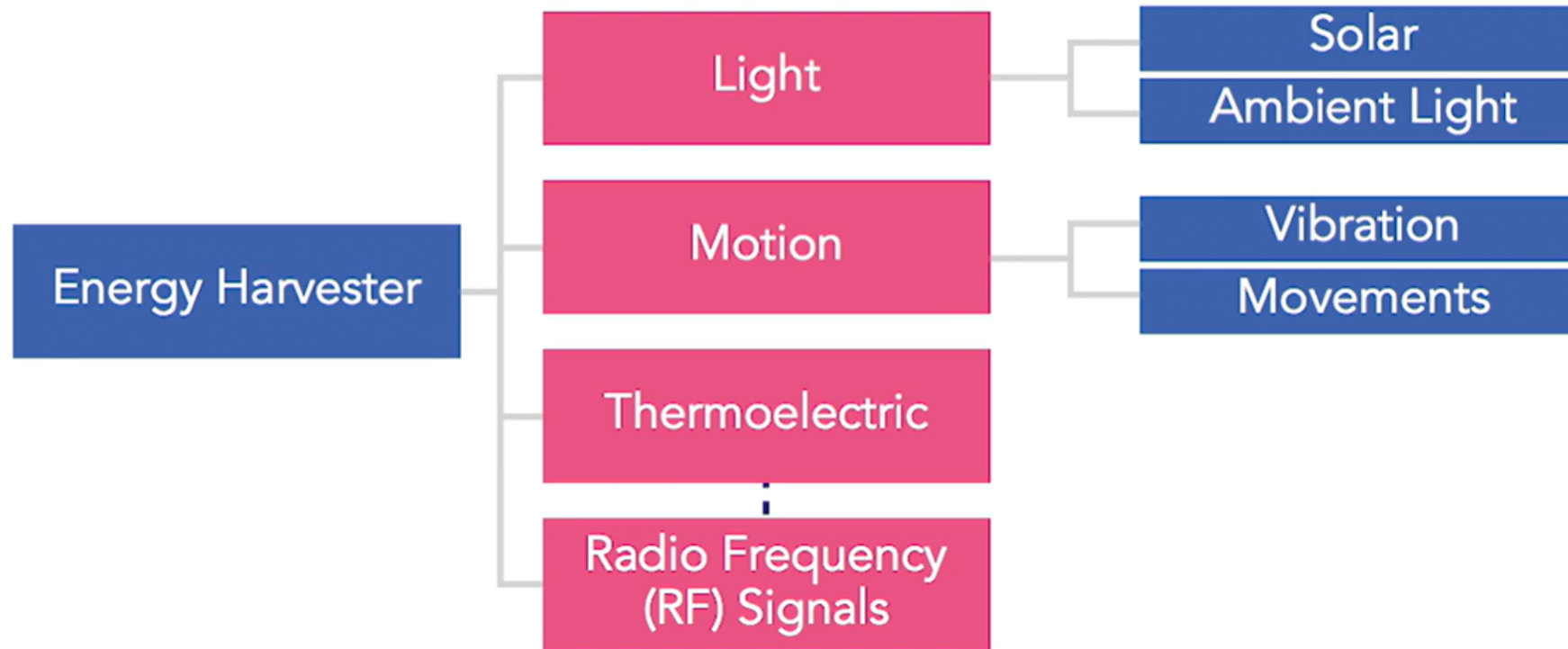
Energy Harvesting Module

- ▶ Some systems convert motion, such as that of ocean waves, into electricity to be used by oceanographic monitoring sensors for autonomous operation. Future applications may include high-power output devices (or arrays of such devices) deployed at remote locations to serve as reliable power stations for large systems. Another application is in wearable electronics, where energy harvesting devices can power or recharge cell phones, mobile computers, and radio communication equipment.
- ▶ Accumulating energy
- ▶ These systems are often very small and require little power, but their applications are limited by the reliance on battery power. Scavenging energy from ambient vibrations, wind, heat, or light could enable smart sensors to function indefinitely.

Energy Harvesting Module

- ▶ Storage of power:
- ▶ energy can be stored in a capacitor, super capacitor, or battery. Capacitors are used when the application needs to provide huge energy spikes. Batteries leak less energy and are therefore used when the device needs to provide a steady flow of energy. These aspects of the battery depend on the type that is used.
- ▶ Use of the power:
- ▶ Current interest in low-power energy harvesting is for independent sensor networks. In these applications, an energy harvesting scheme puts power stored into a capacitor then boosts/regulates it to a second storage capacitor or battery for use in the microprocessor or in the data transmission. The power is usually used in a sensor application and the data is stored or transmitted, possibly through a wireless method.

Energy Harvesting Methods



Examples of Typical EH devices

Light

Temperature and humidity sensor powered by solar cells

Motion

Door handle powered by a motion energy harvester

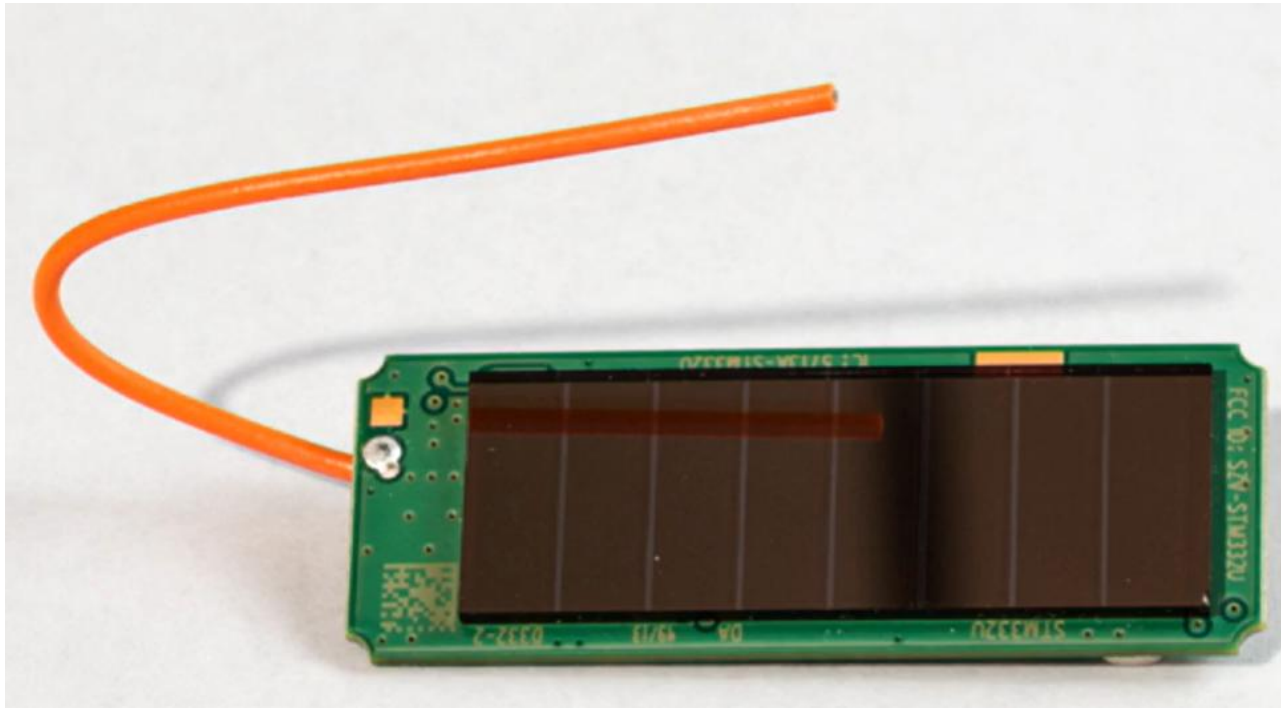
Thermoelectric

Sensor powered by a thermoelectric energy harvester

Radio Frequency (RF) Signals

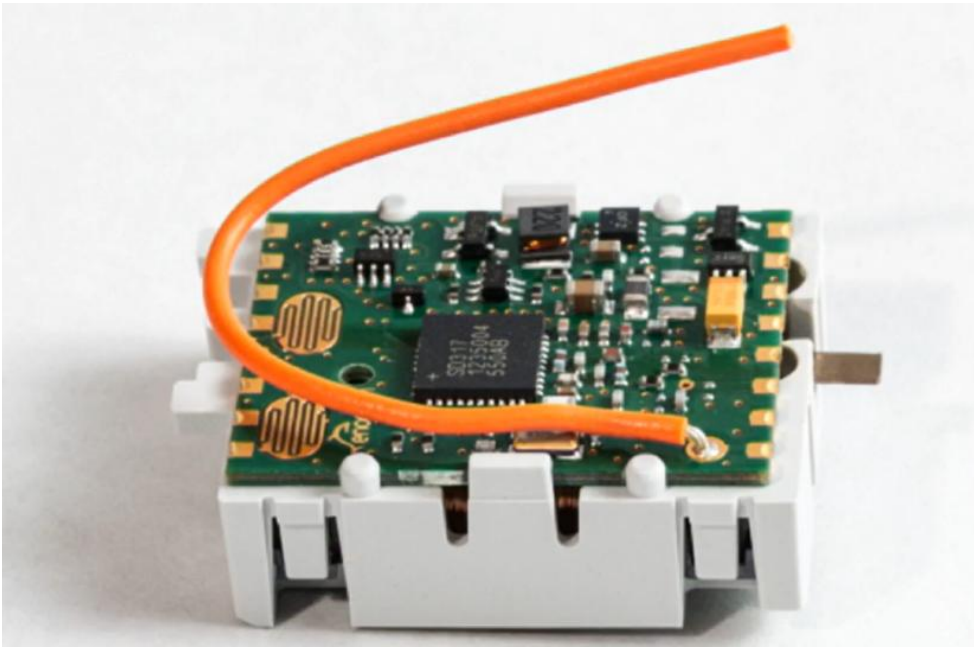
Sensor powered by an RF energy harvester

Solar panel



EnOcen based temperature sensor with a solar cell

EH device



EnOcean Eco 200 Energy harvester consisting of magnets, coils that can generate electric power linear motion with contacts.

Vibration Energy Harvesters

Vibration Energy Harvesters

FEATURES

- Designed to power high performance Wireless Sensor Nodes.
- High power output up to 27.5mW
- High reliability design, including hermetically-sealed stainless steel case.
- Intrinsically Safe for use in Hazardous Areas: ATEX, IECEx, FM and cFM Certifications.
- Stabilized harvesting frequency and bandwidth over full industrial temperature range -40 to +85 °C.
- Two-pin IEC-standard connector.
- Two standard voltage outputs, nominally 5V and 8V.
- Current drive output independent of load up to nominal voltage.
- Center thru-hole for straightforward installation, with other mounting options available.



Vibration Energy Harvester

Vibration based energy harvesters from PMG

1. Performance

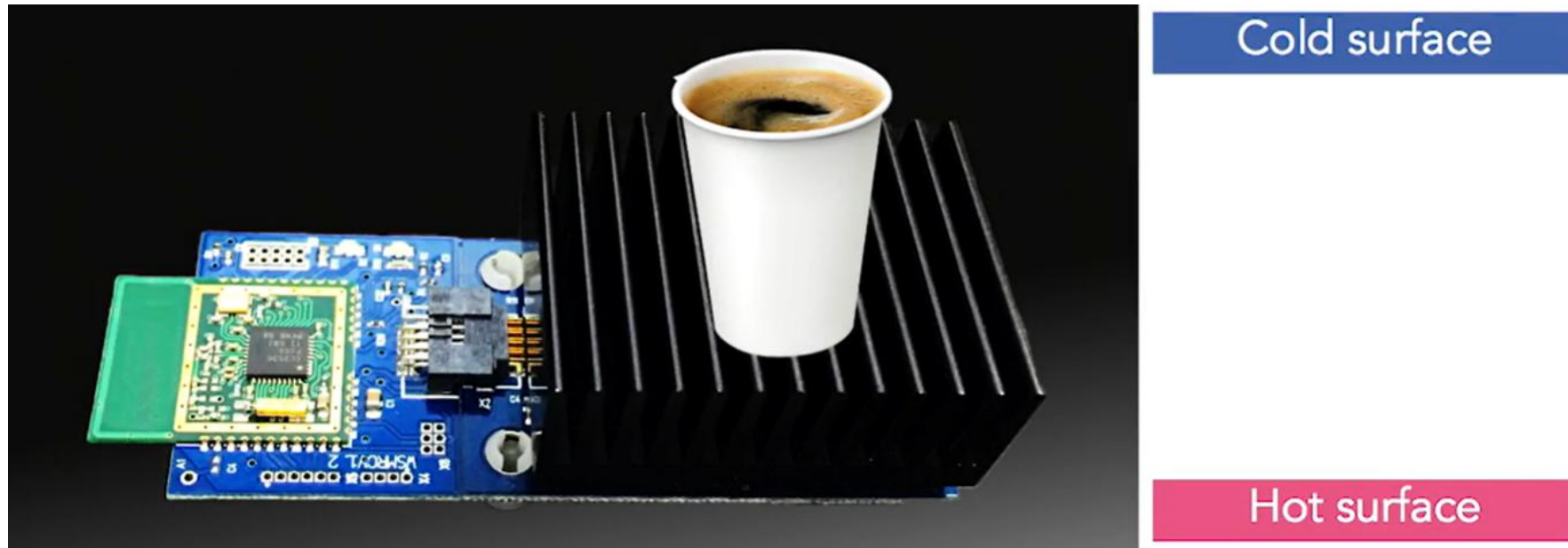
1.1. Power output

	Description	5V Model	8V Model
1.1.1	Maximum	5.5 mA (27.5mW)	3.0 mA (24mW)
1.1.2	Minimum @ 0.050g RMS, nominal frequency	0.840 mA (4.2mW)	0.525 mA (4.2mW)
1.1.3	Minimum @ 0.050g RMS, 2Hz bandwidth	0.600 mA (3mW)	0.375 mA (3mW)
1.1.4	Minimum @ 0.025g RMS, nominal frequency	0.240mA (1.2mW)	0.150mA (1.2mW)

1.2. Physical

	Description	5V and 8V Model	Notes
1.2.1	Temperature range, storage & operation	-40°C TO +85°C	Applies to all parameters

Thermoelectric Energy Harvester system

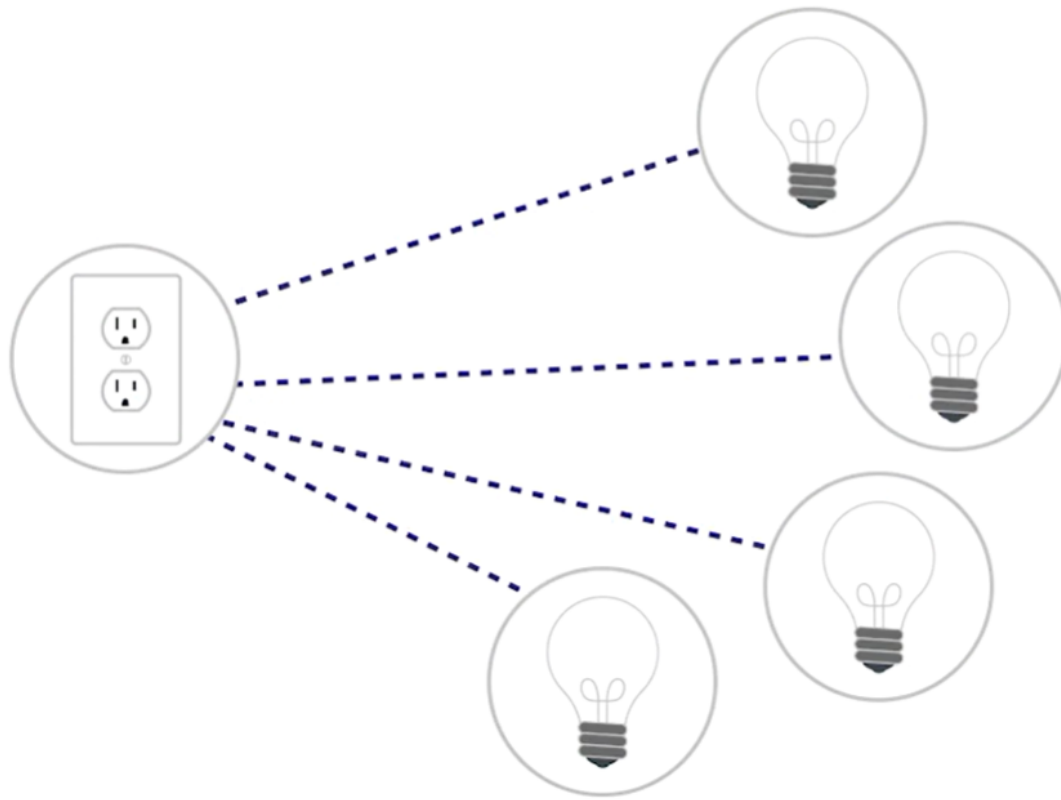


A thermoelectric energy harvesting kit from micropelt is shown here. Electric energy can be generated by the temperature gradient between the surface at the bottom and top

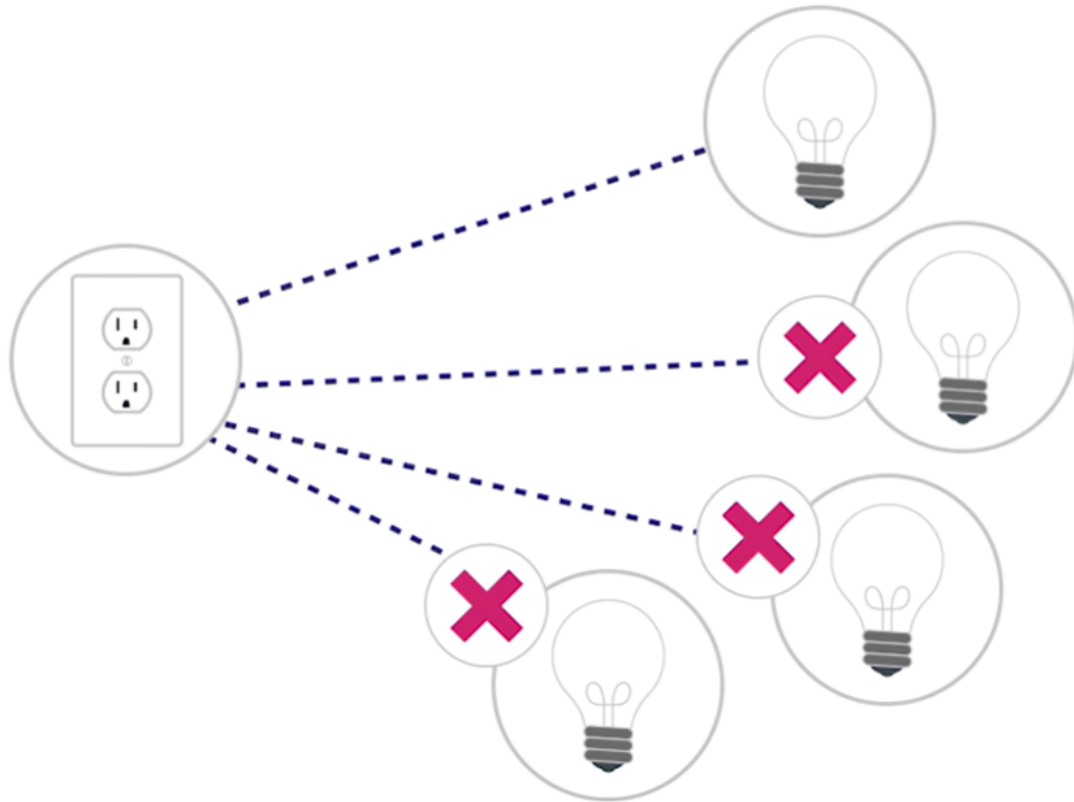
EnOcen features

- ISO/IEC 14543-3-10 radio standard for home/building automation in sub-1 GHz regional unlicensed bands
- EnOcean specifications for radio protocols, equipment profiles, remote management, security, etc.
- Flexible short-range indoor/outdoor communications
- Extremely low-power and short radio telegram transmissions

Data Traffic: Braodcast



Data Traffic: Braodcast



Telegram structure



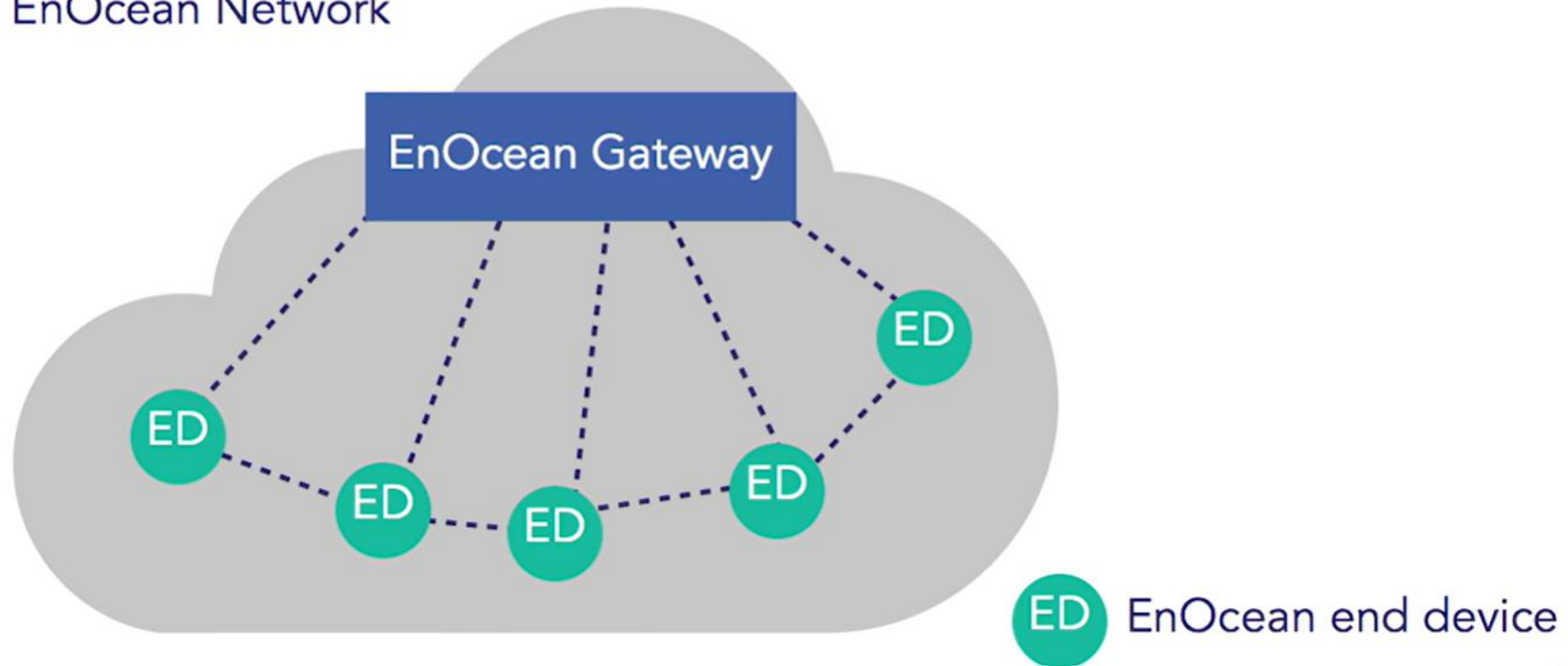
Broadcast telegram structure



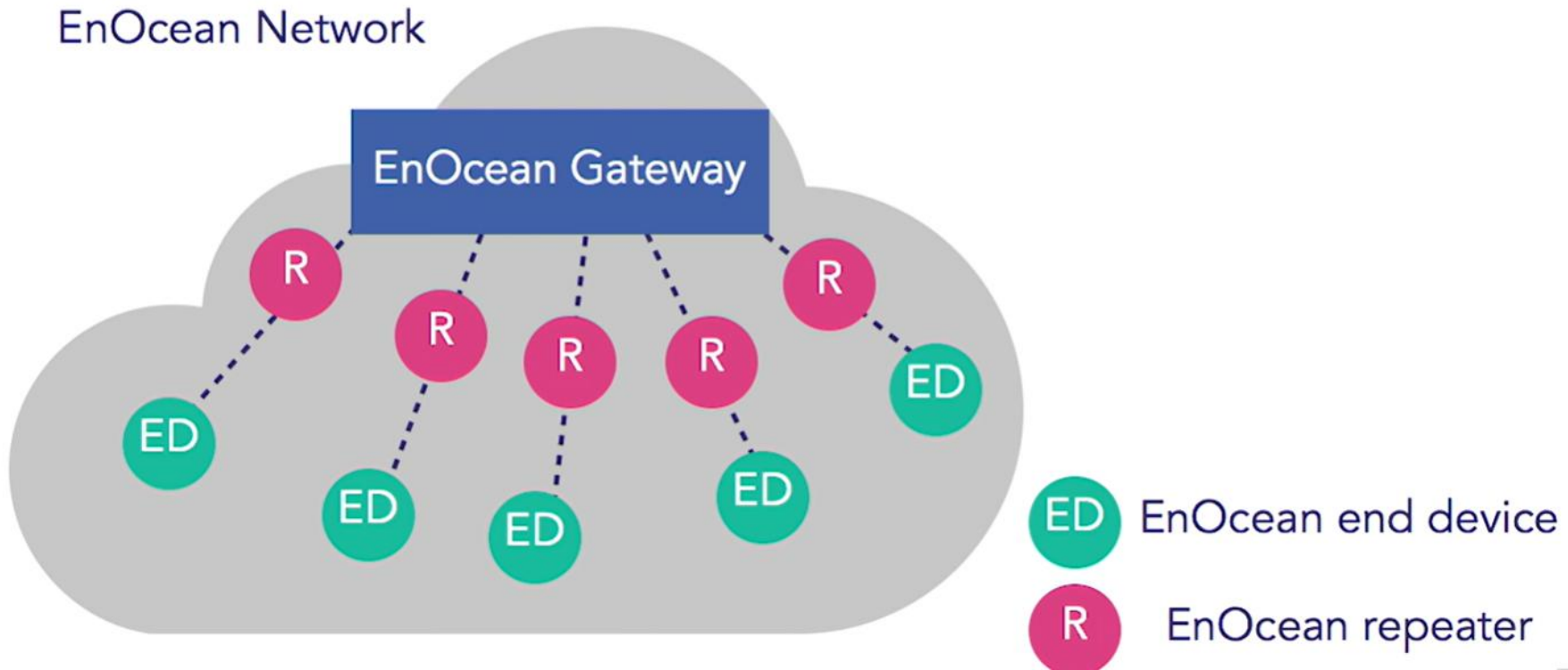
Addressed telegram (ADT) structure

Network Architecture

EnOcean Network

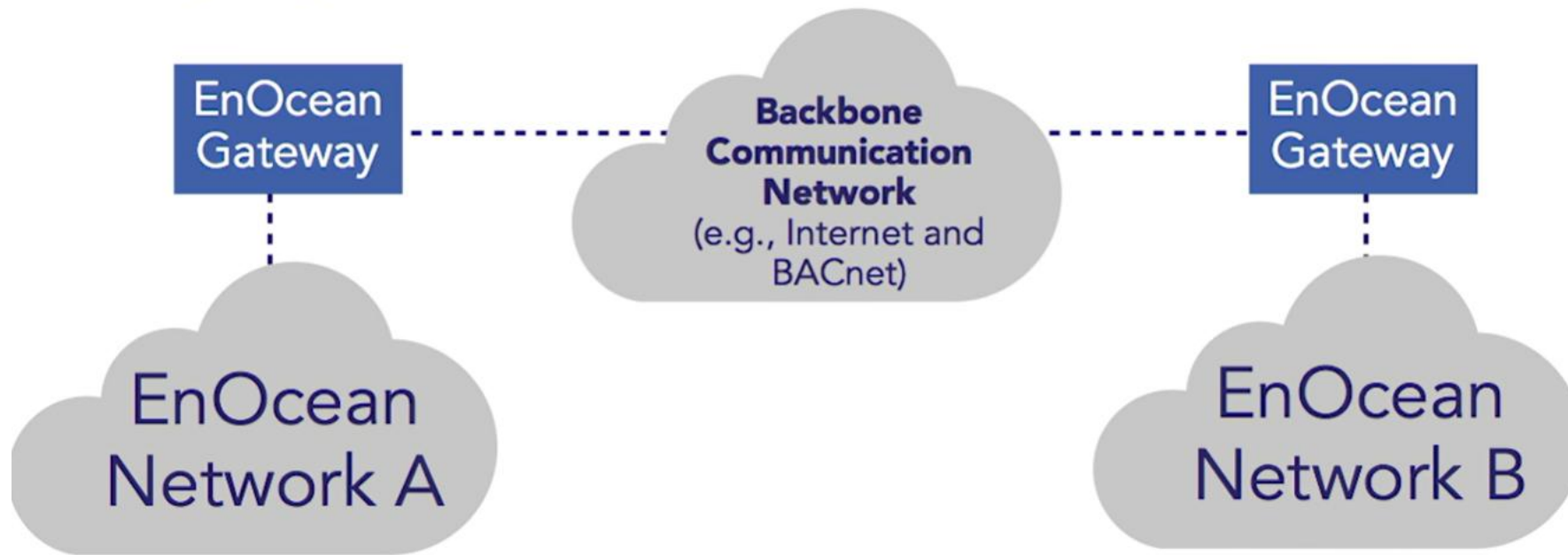


Network Architecture

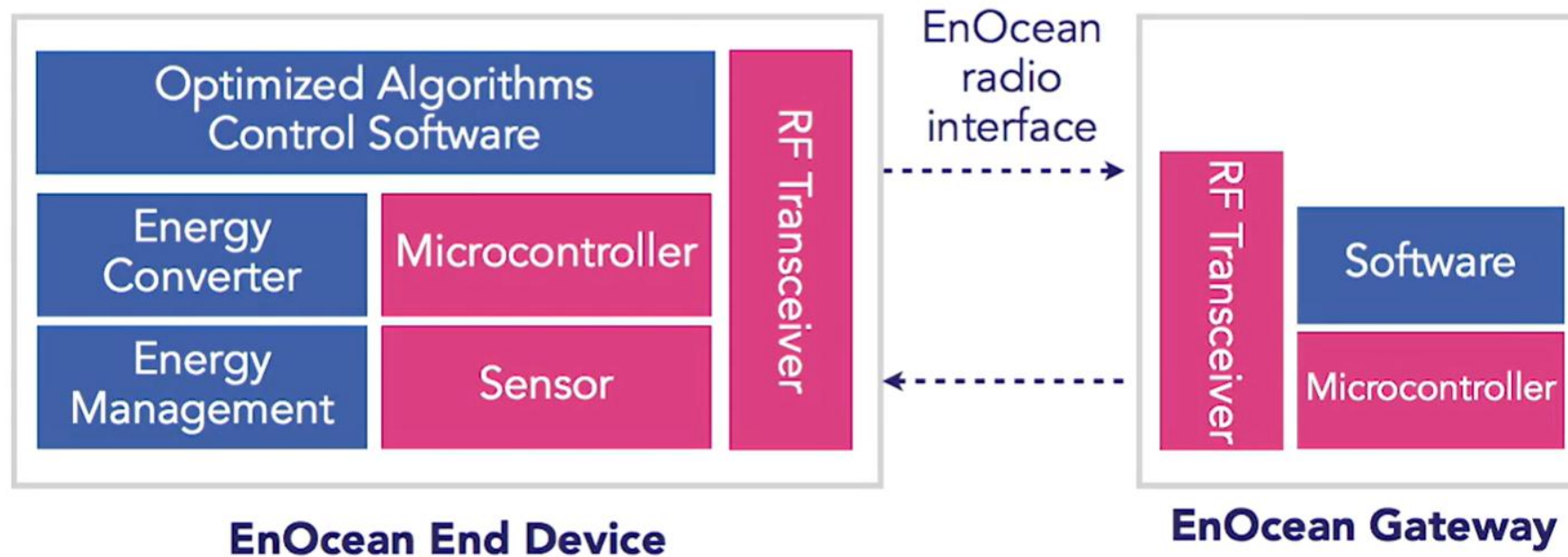


Network Architecture

A long-range communication scenario



Inside an EnOcean System



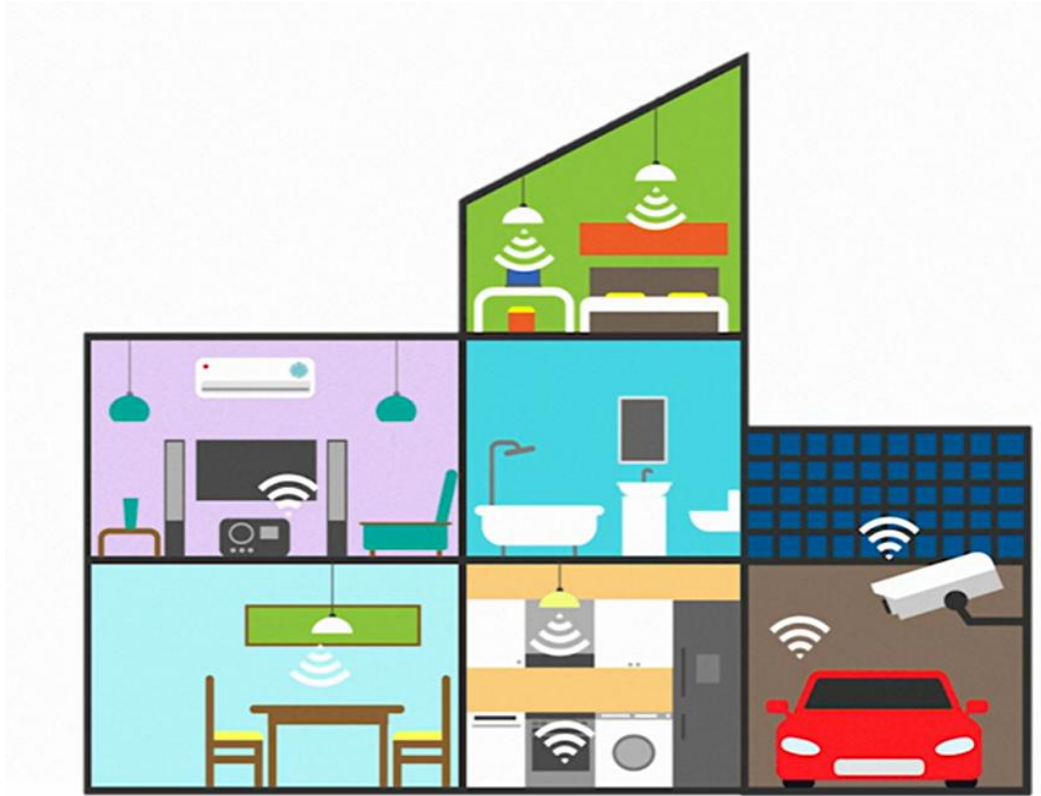
LoWPAN Features

- Consist of low-power, low-cost devices
- Usually offer low bit rates on a communication link
- Operate at unlicensed radio frequency bands
- Support low-power IoT applications

LoWPAN Use cases

- Home automation
- Building automation
- Industrial automation
- Healthcare
- General M2M communications

LoWPAN Use cases



Comparison

	Bluetooth LE	Zigbee	6LoWPAN
Network topology	Star	Star, tree, mesh	Star, tree, mesh
Data rate	Up to 2 Mbps	Up to 250 kbps	Up to 250 kbps
Distance/ range*	50 m (indoor) 140+ m (outdoor)	70-100 m (indoor) 300+ m (outdoor)	Tens of meters or a few hundred meters
Frequency band	ISM 2.4 GHz	ISM 2.4 GHz, 868 MHz (Europe), 915 MHz (US and Australia)	Same as Zigbee

* Actual range depends on the radio configurations and environmental conditions.

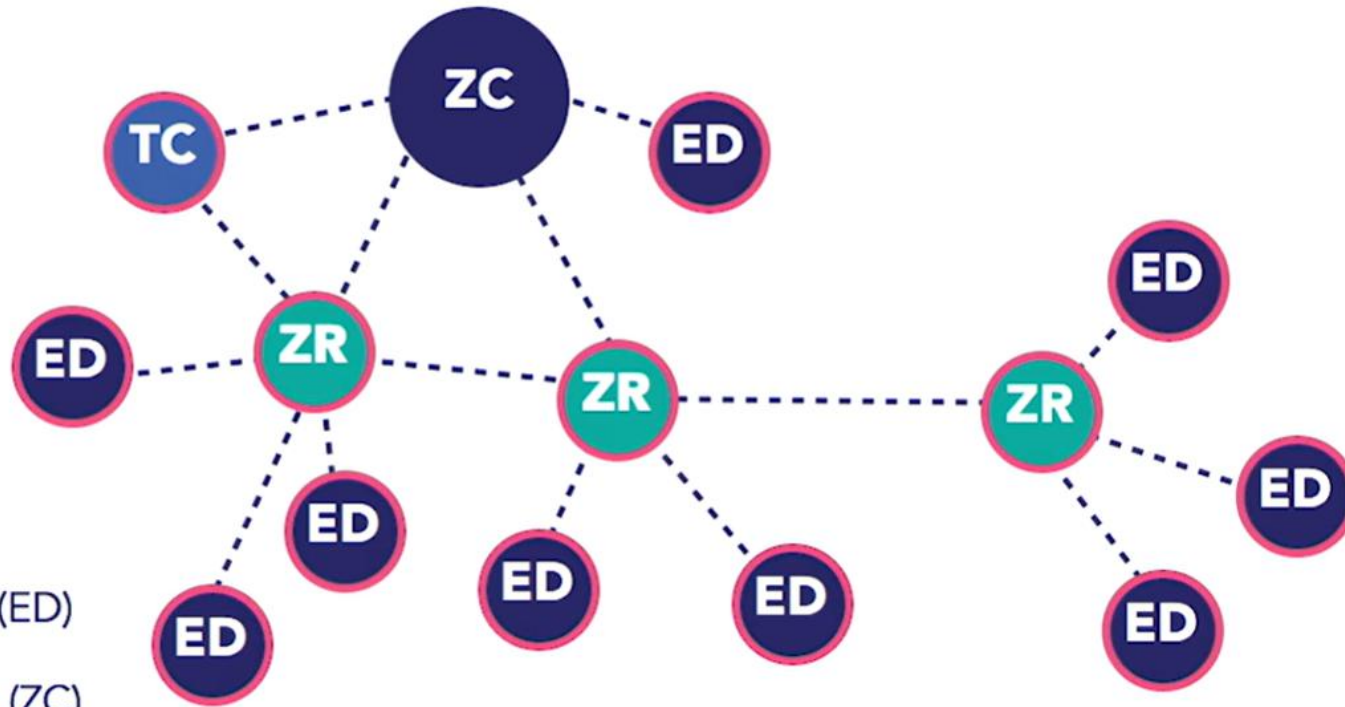
Zigbee Key features

- Worldwide and regional ISM bands: 2.4 GHz frequency band (worldwide), 915 MHz (Americas), and 868 MHz (Europe)
- Support for battery-powered or battery-less end devices
- Support for various network topology options: star, mesh, and tree
- Network security

Zigbee Application

- Environmental control
- Personal healthcare
- Industrial control
- Home automation
- Precision agriculture
- Point-to-point cable replacement

Network Architecture



ZigBee End Device (ED)
ZigBee Router (ZR)
ZigBee Coordinator (ZC)
ZigBee Trust Center (TC)

Zigbee platform

IEEE 802.15.4 hardware platform

ZigBee stack

Provide common operations in ZigBee applications

Network formation, network discovery, management, joining, data transfer, network security, etc.

6LoWPAN features

- Worldwide and regional unlicensed ISM bands
- Various network topology options: star, tree, and mesh
- Support for low-power, resource-constrained end devices
- IEEE 802.15.4-based security schemes
- IPv6 support for IEEE 802.15.4-based wireless networks

IPv6 Support

- Suitable for dense LoWPANs
- Does not require proxies for connecting to other IP networks
- Reusing protocols and tools with IP networks

References

- ▶ 1. IoT - Low power wireless networking

Question



Thank you

